

Spatiotemporal Calibration of Doppler Velocity Logs for Underwater Robots

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Abstract—Accurate fusion of Doppler Velocity Logs (DVLs) with other sensors requires both spatial extrinsics and temporal alignment. However, existing underwater DVL calibration methods often assume known lever arms and synchronized clocks, and rarely estimate full calibration parameters jointly. We propose Unified Iterative Calibration (UIC), a spatiotemporal calibration framework for a DVL and a generic co-sensor that can provide a continuous-time motion estimate. We model the base motion with a continuous-time Gaussian process, enabling velocity queries at time-shifted DVL timestamps. UIC computes a sequential initialization: (i) it estimates the clock offset and DVL scale using low-angular-rate segments via velocity-norm matching, and (ii) estimates the DVL-to-base rotation and lever arm through a covariance-corrected closed-form stacked regression with $SO(3)$ projection. A nonlinear refinement further reduces residuals. Simulations with DVL outliers and pool experiments demonstrate robustness and high accuracy downstream odometry estimation.

I. INTRODUCTION

Doppler Velocity Logs (DVLs) are key sensors for underwater navigation, which can provide body-frame velocity measurements [1]–[5]. For multi-sensor fusion (e.g., DVL+camera/SINS), accurate calibration requires both the spatial extrinsics and temporal alignment, since even a small constant clock offset can cause systematic velocity mismatch under dynamic motion.

Underwater DVL calibration has been studied mainly for DVL–camera systems and for standalone DVLs with navigation-grade reference velocity sources. Westman et al. [6] and Xu et al. [7] formulated DVL–camera calibration as trajectory alignment, but typically rely on known DVL–IMU extrinsics and do not estimate a lever arm and a constant time offset jointly. For IMU-free DVLs, works such as [8]–[11] estimate rotation and scale using SINS/GPS-type velocity references, which limits applicability when the co-sensor does not directly provide body-frame velocities. Temporal calibration for heterogeneous sensors has been widely studied on land by fitting continuous-time trajectories and querying measurements at shifted timestamps (e.g., B-splines/GPs) [12]–[16], but has been less explored for DVLs.

We address DVL spatiotemporal calibration by estimating $\Theta = \{^b\delta_{T,d}, \kappa_d, {}^dR_b, {}^d\mathbf{t}_b\}$, where ${}^b\delta_{T,d}$ is a

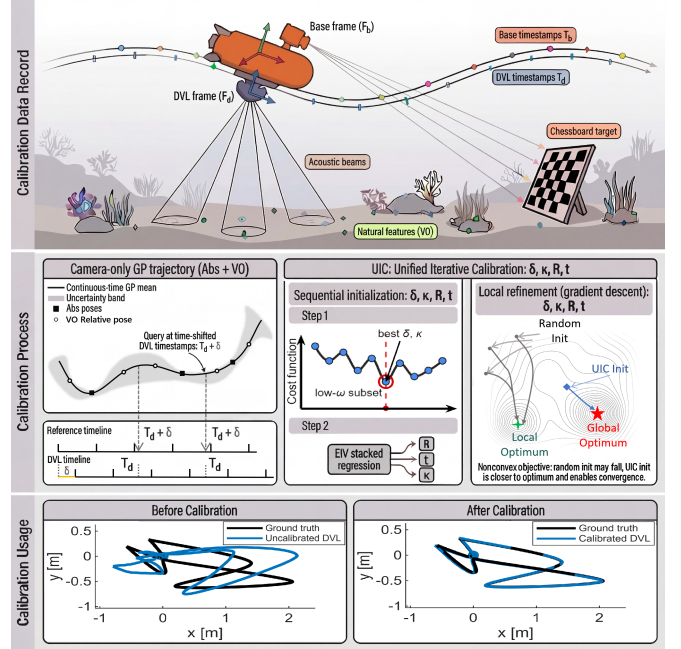


Fig. 1: UIC overview: (1) infer a continuous-time base motion with a GP prior, (2) initialize clock offset/scale using low-rotation segments, (3) Initialize extrinsics via closed-form stacked regression, and (4) optionally refine Θ by local nonlinear least squares.

constant clock offset (base time minus DVL time), κ_d is the DVL velocity scale factor, and $({}^dR_b, {}^d\mathbf{t}_b)$ are the rigid extrinsics. Our formulation does not require the co-sensor to output velocities; it only needs to support a continuous-time motion estimate from which base-frame linear and angular velocities can be queried.

Our main contributions are:

- An EIV spatiotemporal calibration model with large-sample characterization under idealized assumptions. We cast DVL spatiotemporal calibration as an error-in-variables (EIV) nonlinear least-squares problem under a continuous-time GP motion prior, jointly estimating $\Theta = \{^b\delta_{T,d}, \kappa_d, {}^dR_b, {}^d\mathbf{t}_b\}$ via GP-based time-shifted velocity querying. Under idealized modeling assumptions and sufficiently exciting motion, we characterize the large-sample behavior of the proposed sequential estimators, establishing bias elimination for all calibration variables.
- A decoupled calibration pipeline with principled sequential initialization and closed-form extrinsics + numerical refinement. We explicitly decouple the

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clock offset ${}^b\delta_{T,d}$ from the spatial extrinsics to reduce mutual interference in joint optimization: leveraging (i) the rotation-invariance of velocity norms and (ii) near-zero angular-rate segments ($\omega_b \approx 0$) to suppress the lever-arm term, we initialize $({}^b\delta_{T,d}, \kappa_d)$ with minimal dependence on $({}^dR_b, {}^dt_b)$. Fixing the time offset, we then derive a bias-corrected closed-form estimator for the spatial extrinsics (with proper SO(3) projection), and use it as an initialization for optional finite-difference numerical-gradient descent refinement.

- FIM-based excitation analysis and trajectory design guidelines. We analyze the Fisher information matrix (FIM) of the proposed spatiotemporal calibration model to identify the motion excitations required to calibrate all parameters, and translate the analysis into practical trajectory requirements (e.g., combining low-rotation segments for temporal initialization with sufficiently rotational excitation for lever-arm estimation).

Notation: Throughout this paper, we adhere to the following notational conventions: the super-scripted $(\cdot)^o$ represents the true value of a variable $(\cdot)$, $(\cdot)^\circ$ a measurement of $(\cdot)$, and $(\hat{\cdot})$ an estimate to $(\cdot)$. For a vector $a \in \mathbb{R}^3$, let $a^\wedge \in \mathbb{R}^{3 \times 3}$ be the skew-symmetric matrix that satisfies $a \times b = a^\wedge b$ for all $b \in \mathbb{R}^3$, where $\times$ is the vector cross product.

II. Problem formulation

A. DVL Measurement Model

When a DVL is mounted on an underwater robot, it emits acoustic beams and measures the Doppler shifts resulting from the robot's movement relative to its ambient environment, which enables the DVL to calculate a three-dimensional velocity under the assumption of a constant and known acoustic speed.

Mathematically, DVL measurements are resolved relative to its local body-fixed frame $\{\mathcal{F}_d\}$. To analyze the motion of the DVL with respect to a base frame $\{\mathcal{F}_b\}$ (usually attached to the other sensor), let $p_{o_d,b}$ denote the coordinates of the origin o_d of $\{\mathcal{F}_d\}$ expressed in frame $\{\mathcal{F}_b\}$. Essentially, the DVL measures the linear velocity of o_d expressed in its own frame $\{\mathcal{F}_d\}$. Provided that the base frame and the DVL frame are rigidly connected, when the base moves with linear velocity v_b and angular velocity ω_b (both expressed in $\{\mathcal{F}_b\}$), the relationship between the DVL velocity v_d and (v_b, ω_b) is

$$v_d = {}^dR_b(v_b + \omega_b \times {}^dt_b), \quad (1)$$

where ${}^dR_b \in \text{SO}(3)$ is the rotation mapping coordinates from $\{\mathcal{F}_b\}$ to $\{\mathcal{F}_d\}$, and ${}^dt_b \in \mathbb{R}^3$, which equals to $p_{o_d,b}$, is the lever arm from the base-frame origin to the DVL-frame origin expressed in $\{\mathcal{F}_b\}$.

To account for sound-speed mismatch (e.g., due to temperature and salinity) and other multiplicative scale biases, we include a positive scaling factor κ_d . We also

consider an additive measurement noise term. Hence, a real DVL measurement $\hat{v}_d$ can be modeled as

$$\hat{v}_d = \kappa_d {}^dR_b(v_b + \omega_b \times {}^dt_b) + n_v, \quad (2)$$

where $n_v \sim \mathcal{N}(0, Q_d)$ denotes the measurement noise whose covariance can be obtained from the DVL sensor.

B. Unified Spatiotemporal Calibration Problem under GP Motion Prior

The robot motion relative to a fixed world frame is represented by a continuous-time state $x(t)$ expressed in the base frame $\{\mathcal{F}_b\}$. We model $x(t)$ as a Gaussian process (GP),

$$x(t) \sim \mathcal{GP}(\mu(t), K(t, t')), \quad (3)$$

where $K(t, t') \triangleq \text{cov}(x(t), x(t'))$ is the (matrix-valued) GP covariance kernel (with chosen hyperparameters) [17]. The reference sensor suite attached to the base provides measurements $\hat{\mathcal{M}}_b = \{\hat{m}_{b,j}\}_{j=1}^M$ time-stamped at ${}^bT = \{{}^bT_j\}_{j=1}^M$, with measurement model

$$\hat{m}_{b,j} = h(x({}^bT_j)) + n_{b,j}, \quad n_{b,j} \sim \mathcal{N}(0, R_{b,j}), \quad j = 1, \dots, M. \quad (4)$$

Define the stacked vector $\hat{M}_b \triangleq [\hat{m}_{b,1}^\top \dots \hat{m}_{b,M}^\top]^\top$ and let H denote the Jacobian of $h(\cdot)$ evaluated at the linearization point used by GP regression. Let $X_b \triangleq [x({}^bT_1)^\top \dots x({}^bT_M)^\top]^\top$ be the stacked state at the base timestamps. Using $\hat{M}_b$ and the GP prior, we compute a reference trajectory estimate (and its uncertainty) via GP regression, denoted by $\hat{X}_b$ (and Σ_{X_b}); the closed-form expressions are provided in Section III-B.

The DVL provides velocity measurements $\hat{\mathcal{V}}_d = \{\hat{v}_{d,i}\}_{i=1}^N$ time-stamped by its local clock at ${}^dT = \{{}^dT_i\}_{i=1}^N$. A constant clock offset between the base and DVL clocks is modeled by ${}^b\delta_{T,d}$, such that the physical time of a DVL sample recorded at dT_i corresponds to the base time ${}^dT_i + {}^b\delta_{T,d}$. The calibration variables of interest are collected in

$$\Theta \triangleq \{{}^b\delta_{T,d}, {}^dR_b, {}^dt_b, \kappa_d\},$$

where ${}^dR_b \in \text{SO}(3)$ and ${}^dt_b \in \mathbb{R}^3$ are the extrinsic parameters, and $\kappa_d > 0$ is the DVL velocity scale factor. For a candidate ${}^b\delta_{T,d}$, define the aligned DVL timestamps in the base clock by ${}^bT_{d,i}({}^b\delta_{T,d}) \triangleq {}^dT_i + {}^b\delta_{T,d}$, and let $X_d({}^b\delta_{T,d}) \triangleq [x({}^bT_{d,1}({}^b\delta_{T,d}))^\top \dots x({}^bT_{d,N}({}^b\delta_{T,d}))^\top]^\top$ denote the queried states at these aligned instants. Given $\hat{X}_b$, the queried mean $\hat{X}_d({}^b\delta_{T,d})$ is obtained by GP interpolation (as an explicit function of ${}^b\delta_{T,d}$), and the corresponding queried covariance $\Sigma_{X_d}({}^b\delta_{T,d})$ can be computed as a byproduct of GP regression and interpolation following (8) and (9).

At aligned time ${}^bT_{d,i}({}^b\delta_{T,d})$, we expect the DVL measurement as $\kappa_d {}^dR_b(\hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^dt_b)$, where $\hat{v}_{b,i}({}^b\delta_{T,d})$ and $\hat{\omega}_{b,i}({}^b\delta_{T,d})$ are the linear and angular velocity components extracted from the queried state

$\hat{X}_{d,i}(^b\delta_{T,d})$. We then define the DVL least-squares cost

$$J_{d,M}(\hat{X}_d(^b\delta_{T,d}), ^dR_b, ^d\mathbf{t}_b, \kappa_d) \triangleq \sum_{i=1}^N \left\| \hat{v}_{d,i} - \kappa_d ^dR_b (\hat{v}_{b,i}(^b\delta_{T,d}) + \hat{\omega}_{b,i}(^b\delta_{T,d}) \times ^d\mathbf{t}_b) \right\|_{Q_d}^2. \quad (5)$$

We pose spatiotemporal calibration as the following constrained nonlinear least-squares problem:

$$\begin{aligned} \min_{^b\delta_{T,d}, ^dR_b, ^d\mathbf{t}_b, \kappa_d} \quad & J_{d,M}(\hat{X}_d(^b\delta_{T,d}), ^dR_b, ^d\mathbf{t}_b, \kappa_d) \\ \text{s.t.} \quad & ^b\delta_{T,d} \in \mathbb{R}, ^dR_b \in \text{SO}(3), ^d\mathbf{t}_b \in \mathbb{R}^3, \kappa_d > 0. \end{aligned} \quad (6)$$

Problem (6) is an error-in-variables (EIV) problem [18]–[20], because the queried trajectory $\hat{X}_d(^b\delta_{T,d})$ is itself estimated from noisy reference measurements but is independent of the DVL noise. The problem is also highly nonconvex owing to the coupled spatiotemporal parameters and time-shifted GP queries. We therefore develop a sequential estimator of Θ to initialize the subsequent local nonlinear least-squares refinement.

III. METHODOLOGY

UIC consists of sequential initialization followed by local iterative refinement. The initializer estimates all components of Θ without manual or near-true guesses. The reference trajectory $x(t)$ is inferred from the base measurements under the GP prior and is not jointly optimized with Θ , because a flexible trajectory could absorb spatiotemporal miscalibration and confound parameter estimation. Under the idealized EIV assumptions, the initializer is designed to enter a neighborhood of the true parameters, after which refinement further decreases the residual.

A. Sequential Initialization Strategy

Our initialization procedure separates reference motion estimation from the calibration parameters dR_b , $^d\mathbf{t}_b$, κ_d , and $^b\delta_{T,d}$. Concretely, UIC first computes a reference trajectory estimate $\hat{X}_b$ and its uncertainty Σ_{X_b} from the base measurements $\dot{M}_b$ under the GP motion prior. Then, for any candidate clock offset $^b\delta_{T,d}$, the time-shifted motion states at the aligned DVL timestamps are obtained by GP interpolation, yielding $\hat{X}_d(^b\delta_{T,d})$ and $\Sigma_{X_d}(^b\delta_{T,d})$. This stage does not use DVL measurements, so that the reference motion estimate remains independent of the unknown calibration parameters and admits explicit GP-based interpolation. According to [21], the GP regression/interpolation stage can be implemented efficiently and provides both a reference motion estimate and an uncertainty characterization, which are useful for subsequent calibration. The large-sample analysis is derived under idealized modeling assumptions (e.g., correct GP regression/interpolation statistics and sufficiently excited motion), and is intended to justify the design of the initialization estimators rather than to serve as finite-sample performance guarantees.

Directly optimizing Problem (6) from random initialization offers no strong convergence guarantees. Poor initial guesses often lead to divergence or convergence to suboptimal local minima. In our experiments, naive initialization frequently resulted in non-convergence. The initialization procedures are presented in the next two subsections.

B. Initialization of Motion State X

By GP regression conditioned on the base measurements $\dot{M}_b$ and the GP prior, we obtain an estimate $\hat{X}_b$ of the base states X_b in closed form. Then, for any candidate clock offset $^b\delta_{T,d}$, the queried states at the aligned DVL timestamps are computed as $\hat{X}_d(^b\delta_{T,d})$ via standard GP interpolation. Let $^bT_d(^b\delta_{T,d}) \triangleq \{^bT_{d,i}(^b\delta_{T,d})\}_{i=1}^N$, where $^bT_{d,i}(^b\delta_{T,d}) \triangleq ^dT_i + ^b\delta_{T,d}$. By abuse of notation, $K(T_1, T_2)$ denotes the block (cross-)covariance matrix assembled from the GP kernel $K(t, t')$. Define $K_1 \triangleq K(^bT, ^bT)$, $K_2(^b\delta_{T,d}) \triangleq K(^bT, ^bT_d(^b\delta_{T,d}))$, $K_3(^b\delta_{T,d}) \triangleq K(^bT_d(^b\delta_{T,d}), ^bT)$, and $K_4(^b\delta_{T,d}) \triangleq K(^bT_d(^b\delta_{T,d}), ^bT_d(^b\delta_{T,d}))$. Similarly $\mu = \begin{bmatrix} \mu_1 \\ \mu_2(^b\delta_{T,d}) \end{bmatrix}$ which μ_1 denote the GP prior mean at bT and μ_2 denote the GP prior mean at $^bT_d(^b\delta_{T,d})$. According to the linear prediction equations [21, (50)], $\hat{X}_b$ is estimated via GP regression as

$$\hat{X}_b = \mu_1 + K_1 H^\top (R + H K_1 H^\top)^{-1} (M_b - H \mu_1). \quad (7)$$

Then $\hat{X}_d$ is computed as a linear combination of $\hat{X}_b$, with weights determined explicitly by bT , dT , and $^b\delta_{T,d}$, i.e.,

$$\hat{X}_d(^b\delta_{T,d}) = \mu_2(^b\delta_{T,d}) - K_3(^b\delta_{T,d}) K_1^{-1} (\mu_1 - \hat{X}_b). \quad (8)$$

This makes $\hat{X}_d(^b\delta_{T,d})$ an explicit function of the time offset through the GP interpolation weights. In practice, the GP regression in (7) and the query in (8) can be implemented either in closed form for moderate problem sizes, or via sparse/Markov representations when the chosen kernel admits a state-space form; both filtering/smoothing and factor-graph solvers are applicable. In addition, the posterior uncertainty can be computed as a byproduct of GP regression/interpolation [17], given by

$$\begin{aligned} \Sigma_{X_b} &= K_1 - K_1 H^\top (R + H K_1 H^\top)^{-1} H K_1, \\ \Sigma_{X_d}(^b\delta_{T,d}) &= K_4(^b\delta_{T,d}) - \\ &\quad K_3(^b\delta_{T,d}) K_1^{-1} (K_1 - \Sigma_{X_b}) K_1^{-1} K_2(^b\delta_{T,d}). \end{aligned} \quad (9)$$

C. Initialization of Calibration Variables

We next initialize the calibration variables using all DVL measurements together with the GP estimate $\hat{X}_d(^b\delta_{T,d})$. We decouple the calibration of the extrinsic parameters dR_b and $^d\mathbf{t}_b$ from that of the temporal offset and scale parameters $^b\delta_{T,d}$ and κ_d by leveraging specialized robot motion patterns. As will be shown shortly, the resulting estimators can be computed efficiently—via line

search for the temporal/scale parameters, and a closed-form bias-corrected estimator for the spatial extrinsics.

1) Clock offset and scale initialization: In principle, we estimate the clock offset ${}^b\delta_{T,d}$ by maximizing the alignment between the DVL measurements and the predicted DVL output based on $\hat{X}_d({}^b\delta_{T,d})$. Specifically, the predicted DVL velocity is given by $\kappa_d {}^dR_b (\hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b)$ according to (2). To reduce the influence of the spatial extrinsics in this stage, we select a subset of measurements $\mathcal{D}_o$ with small angular velocity, i.e., $\omega_{b,i} \approx 0$. Under this operating regime, the lever-arm term $\hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b$ becomes negligible. Furthermore, taking the Euclidean norm removes the effect of dR_b due to rotation invariance. Combining these yields, for $i \in \mathcal{D}_o$,

$$\kappa_d^2 \|\hat{v}_{b,i}({}^b\delta_{T,d})\|^2 \approx \|v_{d,i}^o\|^2. \quad (10)$$

Centering the noise terms leads to the following two-dimensional search problem:

$$\min_{{}^b\delta_{T,d} \kappa_d} \sum_{i \in \mathcal{D}_o} \left(\frac{\|\hat{v}_{d,i}\|^2 - \text{tr}(Q_d)}{\kappa_d^2} \|\hat{v}_{b,i}({}^b\delta_{T,d})\|^2 + \text{tr}(\Sigma_{v_b,i}({}^b\delta_{T,d})) \right), \quad (11)$$

where $\Sigma_{v_b,i}({}^b\delta_{T,d})$ denotes the 3×3 marginal covariance of the interpolated linear velocity $\hat{v}_{b,i}({}^b\delta_{T,d})$, extracted from $\Sigma_{X_d}({}^b\delta_{T,d})$. The problem can be efficiently solved by performing a grid search over ${}^b\delta_{T,d}$ while varying κ_d over a small grid around 1. Let ${}^b\hat{\delta}_{T,d}$ and $\hat{\kappa}_d$ be any global minimizer. The theorem characterizes large-sample behavior under the idealized regime $\omega_{b,i} = 0$ for $i \in \mathcal{D}_o$.

Remark 1: Under idealized conditions (e.g., sufficiently small angular rates on the selected subset, correct second-moment models for GP interpolation errors, and identifiable speed excitation), the minimizer of (11) is a consistent estimator of $({}^b\delta_{T,d}, \kappa_d)$ as the number of selected samples increases. This remark follows standard EIV-estimation and is included to motivate the estimator structure rather than to provide a finite-sample guarantee.

2) Bias-corrected closed-form extrinsics initialization: Stacking all DVL measurements as $\hat{V}_d = [\hat{v}_{d,1} \cdots \hat{v}_{d,N}]^\top$, we arrive at the stacked version of (2):

$$\hat{V}_d = \kappa_d V_b^{od} R_b^\top - \kappa_d \Omega_b^o ({}^d\mathbf{t}_b)^\top {}^dR_b^\top + N_v, \quad (12)$$

where $V_b^o = [v_{b,1}^o \cdots v_{b,N}^o]^\top$, $\Omega_b^o = [\omega_{b,1}^o \cdots \omega_{b,N}^o]^\top$ and $N_v = [n_{v,1} \cdots n_{v,N}]^\top$. Define

$$Z_1 = \kappa_d {}^dR_b^\top, \quad Z_2 = \kappa_d ({}^d\mathbf{t}_b)^\top {}^dR_b^\top, \quad Z \triangleq \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \in \mathbb{R}^{6 \times 3},$$

and define $H^o \triangleq [V_b^o \quad -\Omega_b^o] \in \mathbb{R}^{N \times 6}$. Then (12) can be written compactly as $\hat{V}_d = H^o Z + N_v$. Omitting the SO(3) constraint of dR_b , we estimate Z by solving

$$\argmin_Z \|\hat{V}_d - H^o Z\|_F^2,$$

which admits the closed-form solution

$$\hat{Z} = \left(\frac{H^{o\top} H^o}{N} \right)^{-1} \left(\frac{H^{o\top} \hat{V}_d}{N} \right). \quad (13)$$

We note that κ_d is estimated again in the above problem. This is a deliberate technical choice: the earlier estimation used only a selected subset $\mathcal{D}_o$ of DVL measurements with $\omega_{b,i} \approx 0$, which represents a relatively limited volume of data. In contrast, the stacked formulation (12) uses the entire measurement set. This yields a reliable estimate of κ_d in the sense that it exhibits less variance around the true value.

The exact values of $v_{b,i}^o$ and $\omega_{b,i}^o$ are not directly measurable, therefore their estimates $\hat{v}_{b,i}$ and $\hat{\omega}_{b,i}$ from $\hat{X}_b$ are substituted into (13). Accordingly, $\hat{Z}$ should be corrected to account for this substitution in the EIV setting. The resulting bias-corrected estimate is

$$\hat{Z}^c = \left(\frac{\hat{H}^\top \hat{H} - G}{N} \right)^{-1} \left(\frac{\hat{H}^\top \hat{V}_d}{N} \right), \quad (14)$$

where $\hat{H} = [\hat{V}_b - \hat{\Omega}_b]$ with $\hat{V}_b = [\hat{v}_{b,1}({}^b\hat{\delta}_{T,d}) \cdots \hat{v}_{b,N}({}^b\hat{\delta}_{T,d})]$ and $\hat{\Omega}_b = [\hat{\omega}_{b,1}({}^b\hat{\delta}_{T,d}) \cdots \hat{\omega}_{b,N}({}^b\hat{\delta}_{T,d})]$, and $G \in \mathbb{R}^{6 \times 6}$ is the covariance-induced correction term. Specifically, letting $\Sigma_{v_b,i}({}^b\hat{\delta}_{T,d})$ and $\Sigma_{\omega_b,i}({}^b\hat{\delta}_{T,d})$ denote the 3×3 marginal covariances of the interpolated linear and angular velocities (extracted from $\Sigma_{X_d}({}^b\hat{\delta}_{T,d})$ in (9)), we define

$$G \triangleq \sum_{i=1}^N \text{blkdiag}(\Sigma_{v_b,i}({}^b\hat{\delta}_{T,d}), \Sigma_{\omega_b,i}({}^b\hat{\delta}_{T,d})).$$

We further reconstruct $\hat{\kappa}_d^c$ from $\hat{Z}^c$ as

$$\hat{\kappa}_d^c = |\det(\hat{Z}_1^c)|^{1/3}, \quad (15)$$

which follows from $\det({}^d\hat{R}_b^o) = 1$. Performing singular value decomposition (SVD) on $(\hat{Z}_1^c / \hat{\kappa}_d^c)^\top$, we write $(\hat{Z}_1^c / \hat{\kappa}_d^c)^\top = U_R \Sigma_R V_R^\top$, and set

$${}^d\hat{R}_b^c = U_R \text{diag}([1, 1, \det(U_R V_R^\top)]) V_R^\top. \quad (16)$$

We then recover an estimate ${}^d\hat{\mathbf{t}}_b^{c\wedge}$ from $\hat{Z}_2^c$ as

$${}^d\hat{\mathbf{t}}_b^{c\wedge} = \left(-\hat{Z}_2^c {}^d\hat{R}_b^c / \hat{\kappa}_d^c - (-\hat{Z}_2^c {}^d\hat{R}_b^c / \hat{\kappa}_d^c)^\top \right) / 2. \quad (17)$$

The following result characterizes the large-sample behavior of the above estimates under standard EIV conditions.

Remark 2: When the motion is sufficiently exciting so that the normal matrix is nonsingular, and when the GP-interpolated velocity errors satisfy standard regularity conditions, the structured estimator recovered from (14)–(17) is consistent under standard EIV arguments. This remark is primarily included to motivate the bias-correction structure; finite-sample behavior is assessed experimentally.

D. Iterative Refinement

Starting from the sequential initialization $\hat{\Theta}$, we refine Θ by local iterative optimization of Problem (6). The rotation is updated on $\text{SO}(3)$ via an exponential-map perturbation, and the DVL scale is re-parameterized to enforce $\kappa_d > 0$:

$${}^dR_b \leftarrow \exp(\phi^\wedge) {}^dR_b, \quad \phi \in \mathbb{R}^3, \quad (18)$$

$$\kappa_d = \exp(s_d), \quad s_d \in \mathbb{R}. \quad (19)$$

We apply numerical gradient descent to refine the parameters.

E. Trajectory Design

The calibration accuracy of Θ depends critically on motion excitation, which can be characterized via a Fisher information matrix (FIM) induced by the DVL velocity residual model in (5). Importantly, the predicted body-frame motion $\hat{v}_b(\cdot)$ and $\hat{\omega}_b(\cdot)$ is obtained from the continuous-time GP and therefore carries uncertainty. To reflect this, we use an effective residual covariance Q_e , which captures not only the intrinsic DVL measurement noise, but also the uncertainty induced by the estimated motion signals $\hat{v}_b, \hat{\omega}_b$.

Define the predicted DVL velocity at aligned time ${}^bT_{d,i}({}^b\delta_{T,d}) = {}^dT_i + {}^b\delta_{T,d}$ as

$$\bar{v}_{d,i}(\Theta) = \kappa_d {}^dR_b \left(\hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b \right).$$

Denote $r_i(\Theta) \triangleq \hat{v}_{d,i} - \bar{v}_{d,i}(\Theta)$ and $y_i({}^b\delta_{T,d}) \triangleq \hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b$. Using the unconstrained scale parameterization $\kappa_d = \exp(s_d)$ and a left perturbation ${}^dR_b \leftarrow \exp(\phi^\wedge) {}^dR_b$, the first-order residual Jacobians satisfy

$$\begin{aligned} J_{s,i} &\triangleq \frac{\partial r_i}{\partial s_d} = -\kappa_d {}^dR_b y_i({}^b\delta_{T,d}), \\ J_{\phi,i} &\triangleq \frac{\partial r_i}{\partial \phi} = \kappa_d ({}^dR_b y_i({}^b\delta_{T,d}))^\wedge, \\ J_{t,i} &\triangleq \frac{\partial r_i}{\partial ({}^d\mathbf{t}_b)} = -\kappa_d {}^dR_b (\hat{\omega}_{b,i}({}^b\delta_{T,d}))^\wedge, \\ J_{\delta,i} &\triangleq \frac{\partial r_i}{\partial {}^b\delta_{T,d}} = -\kappa_d {}^dR_b \left(\dot{\hat{v}}_{b,i}({}^b\delta_{T,d}) + \dot{\hat{\omega}}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b \right), \end{aligned}$$

where $(\dot{\cdot})$ denotes a time derivative evaluated at the aligned instant, which is always available analytically under a continuous-time GP model.

The FIM can be written in the stacked form

$$\mathcal{I}(\Theta) = J^\top Q_e^{-1} J, \quad J \triangleq \begin{bmatrix} J_1 \\ \vdots \\ J_N \end{bmatrix}, \quad J_i \triangleq [J_{\delta,i} \quad J_{s,i} \quad J_{\phi,i} \quad J_{t,i}].$$

Here Q_e is the effective covariance of the residuals, which aggregates the DVL measurement uncertainty and the additional uncertainty caused by motion estimates.

From the FIM, the lever arm ${}^d\mathbf{t}_b$ is excited only by rotation since $J_{t,i} \propto (\hat{\omega}_{b,i}({}^b\delta_{T,d}))^\wedge$; thus $\hat{\omega}_{b,i}({}^b\delta_{T,d}) \approx 0$ makes ${}^d\mathbf{t}_b$ unobservable, and near single-axis rotation yields an ill-conditioned $\mathcal{I}_{tt}$ with the component of ${}^d\mathbf{t}_b$

Algorithm 1 UIC: Sequential Initialization + Gradient-Descent Refinement

Input: time-stamped DVL measurements $\{(\hat{v}_{d,i}, T_{d,i})\}_{i=1}^N$; time-stamped base measurements $\{(\hat{m}_{b,j}, T_{b,j})\}_{j=1}^M$; GP kernel $K(t, t')$ in (3); angular-rate threshold ω_{th} for selecting $\mathcal{D}_o$

Output: ${}^b\hat{\delta}_{T,d}$, $\hat{\kappa}_d$, ${}^d\hat{R}_b$, ${}^d\hat{\mathbf{t}}_b$

- 1: GP reference trajectory: compute $\hat{X}_b(\cdot)$ with covariances from $\{\hat{m}_{b,j}\}$ and K via (7)–(9).
 - 2: Temporal/scale pre-initialization on $\mathcal{D}_o$: solve (11) by a grid search over ${}^b\delta_{T,d}$ and κ_d around 1 only on the low-angular-velocity index set.
 - 3: Bias-corrected stacked extrinsics initialization: using ${}^b\hat{\delta}_{T,d,\text{init}}$, construct $\hat{H}$ and G as in (14), and compute $\hat{Z}^c$ using all $\hat{v}_{d,i}$.
 - 4: Recover scale and extrinsics: compute $\hat{\kappa}_{d,\text{init}}$, ${}^d\hat{R}_{b,\text{init}}$, and ${}^d\hat{\mathbf{t}}_{b,\text{init}}$ from $\hat{Z}^c$ via (15), (16), and (17).
 - 5: Numerical-gradient-descent refinement: perform gradient descent on (6) with respect to Θ . Return the final $\hat{\Theta}$ as $\{{}^b\hat{\delta}_{T,d}, {}^d\hat{R}_b, {}^d\hat{\mathbf{t}}_b, \hat{\kappa}_d\}$.
-

along that axis weakly observable. The clock offset ${}^b\delta_{T,d}$ is mainly informed by translational/rotational accelerations through $J_{b\delta_{T,d,i}}$, so nearly constant-speed and constant-turn-rate motions are uninformative for ${}^b\delta_{T,d}$, while $({}^dR_b, \kappa_d)$ benefit from large, direction-rich $y_i({}^b\delta_{T,d})$. Accordingly, we recommend a multi-regime calibration trajectory: (a) low angular rate with deliberately varying speed/direction to excite ${}^b\delta_{T,d}$ while suppressing lever-arm coupling, and (b) sufficiently rich multi-axis rotation with nonzero translation to excite ${}^d\mathbf{t}_b$ and keep $y_i({}^b\delta_{T,d})$ informative for dR_b and κ_d .

IV. NUMERICAL ANALYSIS

This section presents Monte Carlo simulation results that evaluate the proposed UIC under a camera-DVL calibration problem and quantify how spatiotemporal parameters and motion excitation affect DVL calibration and fusion.

A. Simulation setup and evaluation protocol

Each trial synthesizes a bounded 3D SE(3) trajectory together with three measurement streams: (i) absolute poses (Abs) available only in a “calibration zone” and under low-speed/low-turn-rate gating, simulating camera visualizing a calibration board. (ii) relative poses (VO) at a fixed rate simulating visual odometry, and (iii) DVL body-frame linear velocities. The ground-truth calibration parameters $\Theta^o = \{\delta^o, \kappa_d^o, {}^dR_b^o, {}^d\mathbf{t}_b^o\}$ are fixed across MC runs.

B. UIC against robust baselines

We evaluate both calibration accuracy and downstream trajectory impact in this simulation. Calibration accuracy: Over $n_{\text{MC}} = 50$ trials, we estimate $\hat{\Theta}$ and report the errors in $\delta[\text{s}]$, κ_d , ${}^dR_b[^\circ]$, and ${}^d\mathbf{t}_b[\text{m}]$. Trajectory

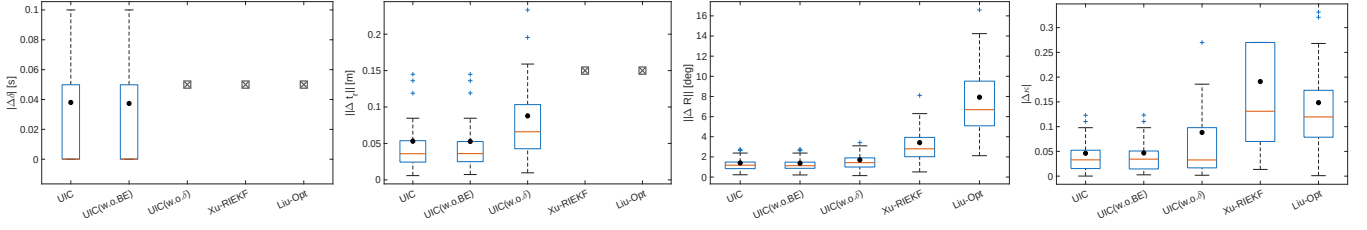


Fig. 2: Calibration Results (Absolute Error distributions of $\Delta\delta$, Δt , ΔR , and $\Delta\kappa$) and RMSE. Boxes show the interquartile range with the median; whiskers indicate the non-outlier range; black dots denote the RMSE.

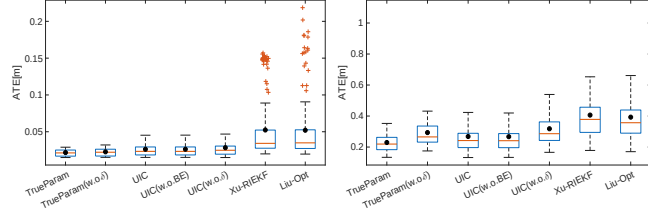


Fig. 3: ATE for simple(left) and aggressive(right) test trajectories

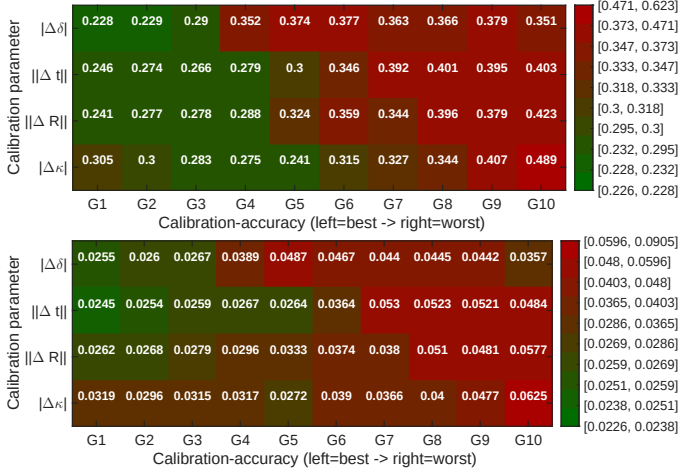


Fig. 4: Calibration accuracy vs Simple(Upper) and Aggressive(Lower) trajectory ATE(Upper)

impact: Each calibrated parameter set is evaluated on two types of independent test trajectories, each of which contains 20 different trajectories. The first type of test trajectories (Simple Trajectories) has slow motions, while the other (Aggressive Trajectories) has rapid movements. We run a GP-InEKF and RTS fusion that incorporates Abs/VO and DVL (with discrete-time offset compensation), and compute the ATE.

To quantify the practical gain of jointly calibrating time and space, we benchmark UIC against two reproduced robust calibration baselines for DVL, Xu-RIEKF [8] and Liu-Opt [10], which estimate only (dR_b, κ_d) ; thus, following their intended usage in our setting, we fix the clock offset δ and lever arm d_t to measured values when running them. We also in-

clude two UIC ablations, UIC(w.o.BE) (without bias-elimination in initialization) and UIC(w.o. δ) (disabling temporal calibration by fixing $\delta = 0$), under the same protocol; Figs. 2 and 3 summarize calibration errors and the downstream ATE, respectively. In addition, to explicitly isolate the impact of temporal misalignment on trajectory estimation, we report the baseline TrueParam and TrueParam(w.o. δ), where the spatial parameters (dR_b, d_t, κ_d) are set to their ground-truth values while δ is set to real value and zero; therefore, any ATE degradation observed in TrueParam(w.o. δ) can be attributed to the wrong time alignment rather than imperfect spatial calibration.

Overall, the results show that explicitly estimating a constant δ together with d_t is a key factor for robust and accurate calibration: when δ is fixed, the remaining parameters (dR_b, κ_d, d_t) tend to absorb spatiotemporal mismatch, which degrades both calibration quality and trajectory accuracy, whereas UIC yields consistently better calibration and ATE, with the advantage being most pronounced under dynamically rich motions where temporal misalignment is more damaging; consistently, TrueParam(w.o. δ) shows only limited ATE change on low-dynamic (Simple) trajectories but a clear degradation on high-dynamic (Aggressive) trajectories, confirming that accurate temporal calibration becomes critical when linear/angular velocities are large. The bias-elimination option provides a modest but consistent improvement for spatial parameters, but its impact is smaller than that of enabling temporal calibration and lever-arm estimation, indicating that these two components dominate this benchmark. Finally, the stratified analysis in Figs. 4 confirms that trajectory accuracy deteriorates as calibration quality worsens, and that the trajectory is generally more sensitive to the clock offset δ than to the scale κ_d .

C. Angular velocity excitation ablation with fixed time offset

This experiment isolates the effect of angular velocity excitation on the spatial DVL-IMU calibration, with an emphasis on the lever arm d_t . From section III-E, when angular motion is weak, the lever arm becomes weakly observable. To decouple spatial calibration from temporal misalignment, we fix the time offset to the

Case	T [s]	$\ \omega_b\ $ [rad/s]	$\Delta\kappa_d$	ΔR [deg]	Δt [m]	runtime [s]
HIGH_T100	100	0.602	0.0683	1.563	0.0976	0.510
LOW_T100	100	0.381	0.0596	1.567	0.1904	0.507
LOW_T2000	2000	0.393	0.0900	0.530	0.1325	18.35

TABLE I: Summary for the angular velocity excitation ablation.

ground truth and disable its estimation in UIC, i.e., $\delta = \delta^o = 0.05$ s. Therefore the time-offset error is identically zero ($\text{RMSE}(\Delta\delta) = 0$) by construction, and UIC estimates only the DVL scale factor, rotation, and lever arm.

We compare three trajectory scenarios over $n_{\text{MC}} = 50$ Monte Carlo runs: (i) HIGH_T100: high angular velocity excitation, $T = 100$ s; (ii) LOW_T100: low angular velocity excitation, $T = 100$ s; (iii) LOW_T2000: low angular velocity excitation, $T = 2000$ s. As a scalar proxy for excitation strength, we report the mean ground-truth angular-speed magnitude $\|\omega_b\|$.

Table I show that lever-arm accuracy is strongly excitation-limited. For the same duration $T = 100$ s, increasing angular-rate excitation reduces the lever-arm RMSE. In contrast, the rotation error remains around ≈ 1.56 deg for both 100 s cases, suggesting that in this regime dR_b is less sensitive to short-horizon excitation amplitude than ${}^d\mathbf{t}_b$. Extending a low-excitation trajectory to $T = 2000$ s improves the rotation error substantially and also improves lever-arm accuracy, but it increases computation time markedly.

V. Real-World Experiment

We conducted a real-world experiment to evaluate our calibration method. We recorded our calibration trajectories in a $4.5 \times 10 \times 2$ m pool. An A50 DVL (around 10 Hz) is mounted on an underwater robot together with a forward-looking stereo camera (10 Hz). Fig. 6 shows the setting of our experiment. The stereo camera provides absolute pose measurements by observing an AprilTag board affixed to the pool wall. We calibrate the camera-DVL extrinsics ${}^dR_c, {}^d\mathbf{t}_c$, their clock offset ${}^c\delta_{T,d}$, and the DVL scale factor κ_d .

We apply GP with WNOJ [22] prior to estimating trajectories and manually set the kernel Q_c as $\text{diag}([100, 100, 100, 100, 100, 100])$. To evaluate the calibration performance, we construct an odometry that integrates the DVL linear velocity measurements with the stereo camera’s absolute orientation measurements as follows. Let ${}^wR_c(T_c)$ denote the camera orientation in the world frame at time T_c . Over a short interval $[T_{c,1}, T_{c,2}]$ between two camera measurements, since the DVL has the same frequency as the camera, there exist two DVL measurements $\hat{v}_{d,1}$ and $\hat{v}_{d,2}$ at times $T_{d,1} + {}^c\delta_{T,d}$ and $T_{d,2} + {}^c\delta_{T,d}$, which satisfy $T_{d,1} + {}^c\delta_{T,d} \leq T_{c,1}$ and $T_{d,2} + {}^c\delta_{T,d} \in [T_{c,1}, T_{c,2}]$. Then, the average velocity in this interval can be approximated

by $v_d = \frac{T_{d,2} + {}^c\delta_{T,d} - T_{c,1}}{\Delta_T} \hat{v}_{d,1} + \frac{T_{c,2} - T_{d,2} - {}^c\delta_{T,d}}{\Delta_T} \hat{v}_{d,2}$, where $\Delta_T = T_{c,2} - T_{c,1}$. Finally, the world-frame camera position update over this interval is approximated by ${}^w\mathbf{p}_2 \approx {}^w\mathbf{p}_1 + \kappa_d^{-1} {}^wR_c(T_{c,1}) {}^cR_d v_d \Delta_T + ({}^wR_c(T_{c,1}) - {}^wR_c(T_{c,2})) {}^d\mathbf{t}_c$, where $\kappa_d^{-1} {}^wR_c(T_{c,1}) {}^cR_d v_d \Delta_T$ and $({}^wR_c(T_{c,1}) - {}^wR_c(T_{c,2})) {}^d\mathbf{t}_c$ denote the updates brought by robot translation and rotation, respectively.

We compare the above estimated position-only trajectory against the reference trajectory derived from the stereo camera’s absolute-position measurements from the AprilTag board. Absolute Trajectory Error (ATE) and Relative Position Error (RPE) are selected as evaluation metrics. We evaluate five methods and compare their performance:

- UIC: our UIC method in Algorithm 1.
- GD (init. 1): gradient decent over Problem (6) initialized with CAD-derived extrinsic parameters.
- GD (init. 2): gradient decent over Problem (6) initialized with ${}^dR_b = \mathbf{I}_3$, ${}^d\mathbf{t}_b = \mathbf{0}$.
- Manual w. δ_T : use manually measured dR_b and ${}^d\mathbf{t}_b$ obtained from the CAD assembly model, and compensate for the clock offset estimated by UIC.
- Manual w.o. δ_T : use manually measured dR_b and ${}^d\mathbf{t}_b$ obtained from the CAD assembly model, with the clock offset fixed at 0.

We recorded three mild-motion trajectories, denoted as A, B, and C, under the same pool and camera-visibility constraints. Each trajectory is used for calibration and evaluated on the other two, yielding six cross-trajectory pairs; the results are summarized in Table II, with representative trajectories shown in Fig. 5. Trajectory C is generally more challenging for the downstream odometry: nearly all methods show larger ATE when evaluated on C, while UIC remains comparable to well-initialized GD and achieves the best average ATE and RPE without manual initialization. Poorly initialized GD is unstable, and UIC outperforms both manual base-lines. The small gap between manual calibration with and without time-offset compensation is consistent with the mild motions; simulations show a larger temporal effect under aggressive dynamics.

Limitations: The lever arm ${}^d\mathbf{t}_c$ is weakly observable in the pool experiment because keeping the AprilTag board within the camera’s field of view limits rotational excitation; consequently, the $\omega \times {}^d\mathbf{t}_c$ contribution in Fig. 7 is negligible. This is primarily an excitation limitation: richer multi-axis rotation, enabled by a wider-view or multi-target reference setup, is the preferred remedy. Priors may stabilize estimation but cannot replace missing excitation.

VI. CONCLUSION

In this paper, we formalized DVL spatiotemporal calibration as an error-in-variables nonlinear least-squares problem under a continuous-time GP motion model, and introduced a UIC framework centered on a sequential,

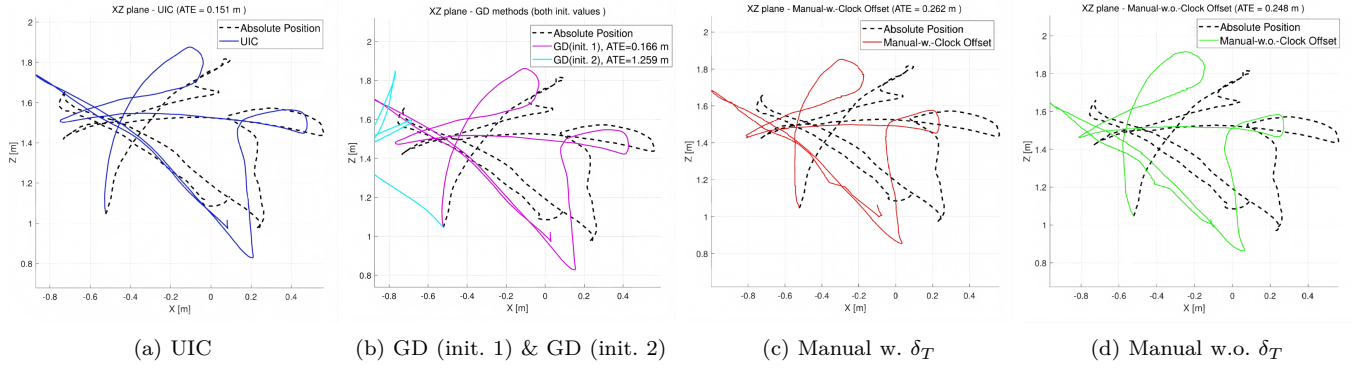


Fig. 5: Trajectory A estimation using the parameters calibrated from trajectory B.

TABLE II: Cross-trajectory ATE and RPE. Each column pair lists the evaluation trajectory under the calibration trajectory shown above. **Bold blue** and **light blue** denote the best and second-best results. Averages are arithmetic means over the six pairs; percentages are relative to UIC and are computed from the listed per-pair values before rounding the averages.

Calibration		Traj A				Traj B				Traj C				Average	
Evaluate		Traj B		Traj C		Traj A		Traj C		Traj A		Traj B		Comparison	
Method	Result	ATE	RPE	ATE	RPE	ATE	RPE	ATE	RPE	ATE	RPE	ATE	RPE	ATE	RPE
UIC	Meter	0.348	0.107	0.621	0.156	0.151	0.070	0.538	0.141	0.209	0.067	0.230	0.071	0.350	0.102
GD	Meter	0.368	0.107	0.606	0.154	0.166	0.070	0.529	0.142	0.231	0.070	0.237	0.072	0.356	0.103
(init. 1)	%	105.7	100	97.58	98.72	109.9	100	98.3	100.71	110.5	104.5	103.0	101.4	101.9	100.5
GD	Meter	0.367	0.107	0.993	0.144	1.259	0.201	0.942	0.138	1.158	0.196	0.750	0.102	0.912	0.148
(init. 2)	%	105.5	100	159.9	92.31	833.8	287.1	175.1	97.9	554.1	292.5	326.1	143.7	260.8	145.1
Manual	Meter	0.378	0.108	0.621	0.158	0.262	0.085	0.607	0.157	0.262	0.085	0.390	0.109	0.420	0.117
(w. δ_T)	%	108.6	100.9	100	101.3	173.5	121.4	112.8	111.3	125.4	126.9	169.6	153.5	120.2	114.7
Manual	Meter	0.388	0.108	0.632	0.158	0.248	0.088	0.632	0.158	0.248	0.088	0.388	0.108	0.423	0.118
(w.o. δ_T)	%	111.5	100.9	101.8	101.3	164.2	125.7	117.5	112.1	118.7	131.3	168.7	152.1	120.9	115.7

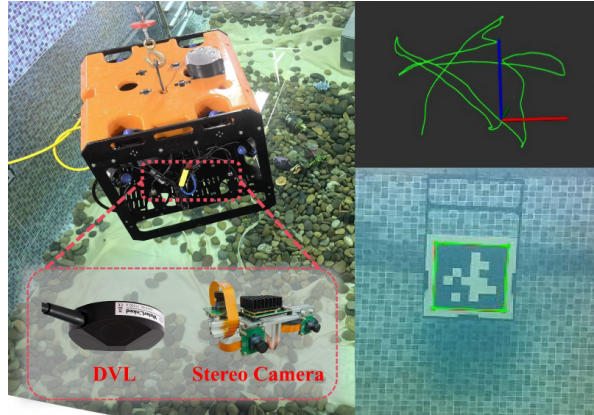


Fig. 6: Water pool experiments. Left: Underwater robot; bottom right: AprilTag board; upper right: robot trajectory.

data-driven initialization procedure with gradient descent refinement. Under idealized conditions, we characterized the large-sample behavior of the proposed initialization estimators. Our simulations show that lever-arm estimation is excitation-limited, and the pool experiment validates the practicality of the proposed initializer under our platform constraints. Monte Carlo ablation studies further highlight that explicitly calibrating the clock offset and executing sufficiently rotationally exciting trajectories are critical for reliable spatiotemporal cal-

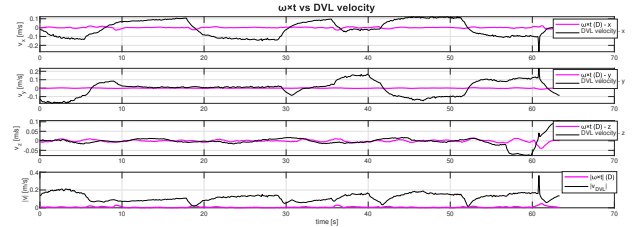


Fig. 7: DVL velocity versus the lever-arm-induced term $\omega \times d_c$.

ibration in finite data.

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References

- [1] L. Zhao, M. Zhou, and B. Loose, “Tightly-coupled visual-dvl- inertial odometry for robot-based ice-water boundary exploration,” in 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023, pp. 7127–7134.
- [2] Y. Huang, P. Li, S. Yan, Y. Ou, Z. Wu, M. Tan, and J. Yu, “Tightly-coupled visual-dvl fusion for accurate localization of underwater robots,” in 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023, pp. 8090–8095.

- [3] S. Xu, K. Zhang, Z. Hong, Y. Liu, and S. Wang, "Diso: Direct imaging sonar odometry," in 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, pp. 8573–8579.
- [4] J. McConnell, F. Chen, and B. Englot, "Overhead image factors for underwater sonar-based slam," IEEE Robotics and Automation Letters, vol. 7, no. 2, pp. 4901–4908, 2022.
- [5] S. E. Webster, L. L. Whitcomb, and R. M. Eustice, "Preliminary results in decentralized estimation for single-beacon acoustic underwater navigation," in Robotics: Science and Systems, vol. 10, 2010.
- [6] E. Westman and M. Kaess, "Underwater apriltag slam and calibration for high precision robot localization," in tech. rep. Carnegie Mellon University Pittsburgh, PA, 2018.
- [7] S. Xu, T. Luczynski, J. S. Willners, Z. Hong, K. Zhang, Y. R. Petillot, and S. Wang, "Underwater visual acoustic slam with extrinsic calibration," in 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2021, pp. 7647–7652.
- [8] B. Xu and Y. Guo, "A novel dvl calibration method based on robust invariant extended kalman filter," IEEE Transactions on Vehicular Technology, vol. 71, no. 9, pp. 9422–9434, 2022.
- [9] B. Xu, L. Wang, S. Li, and J. Zhang, "A novel calibration method of sins/dvl integration navigation system based on quaternion," IEEE Sensors Journal, vol. 20, no. 16, pp. 9567–9580, 2020.
- [10] S. Liu, T. Zhang, and Y. Zhu, "A gnss aided calibration method for dvl error based on the optimal-request," IEEE Sensors Journal, vol. 22, no. 22, pp. 21 899–21 910, 2022.
- [11] B. Wang, J. Liu, Z. Deng, and M. Fu, "A model-free calibration method of inertial navigation system and doppler sensors," IEEE Sensors Journal, vol. 21, no. 2, pp. 2219–2229, 2021.
- [12] P. Furgale, J. Rehder, and R. Siegwart, "Unified temporal and spatial calibration for multi-sensor systems," in 2013 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2013, pp. 1280–1286.
- [13] T. Qin and S. Shen, "Online temporal calibration for monocular visual-inertial systems," in 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2018, pp. 3662–3669.
- [14] J. Peršić, L. Petrović, I. Marković, and I. Petrović, "Spatiotemporal multisensor calibration via gaussian processes moving target tracking," IEEE Transactions on Robotics, vol. 37, no. 5, pp. 1401–1415, 2021.
- [15] M. Li and A. I. Mourikis, "3-d motion estimation and online temporal calibration for camera-imu systems," in 2013 IEEE International Conference on Robotics and Automation, 2013, pp. 5709–5716.
- [16] J. Rehder, R. Siegwart, and P. Furgale, "A general approach to spatiotemporal calibration in multisensor systems," IEEE Transactions on Robotics, vol. 32, no. 2, pp. 383–398, 2016.
- [17] C. K. Williams and C. E. Rasmussen, Gaussian processes for machine learning. MIT press Cambridge, MA, 2006, vol. 2, no. 3.
- [18] A. Wald, "The fitting of straight lines if both variables are subject to error," The annals of mathematical statistics, vol. 11, no. 3, pp. 284–300, 1940.
- [19] B.-Q. Mu and H.-F. Chen, "Recursive identification of multi-input multi-output errors-in-variables hammerstein systems," IEEE Transactions on Automatic Control, vol. 60, no. 3, pp. 843–849, 2014.
- [20] O. Reiersøl, "Identifiability of a linear relation between variables which are subject to error," Econometrica: Journal of the Econometric Society, pp. 375–389, 1950.
- [21] C. H. Tong, P. Furgale, and T. D. Barfoot, "Gaussian process gauss-newton for non-parametric simultaneous localization and mapping," The International Journal of Robotics Research, vol. 32, no. 5, pp. 507–525, 2013. [Online]. Available: <https://doi.org/10.1177/0278364913478672>
- [22] T. Y. Tang, D. J. Yoon, and T. D. Barfoot, "A white-noise-on-jerk motion prior for continuous-time trajectory estimation on $se(3)$," IEEE Robotics and Automation Letters, vol. 4, no. 2, pp. 594–601, 2019.